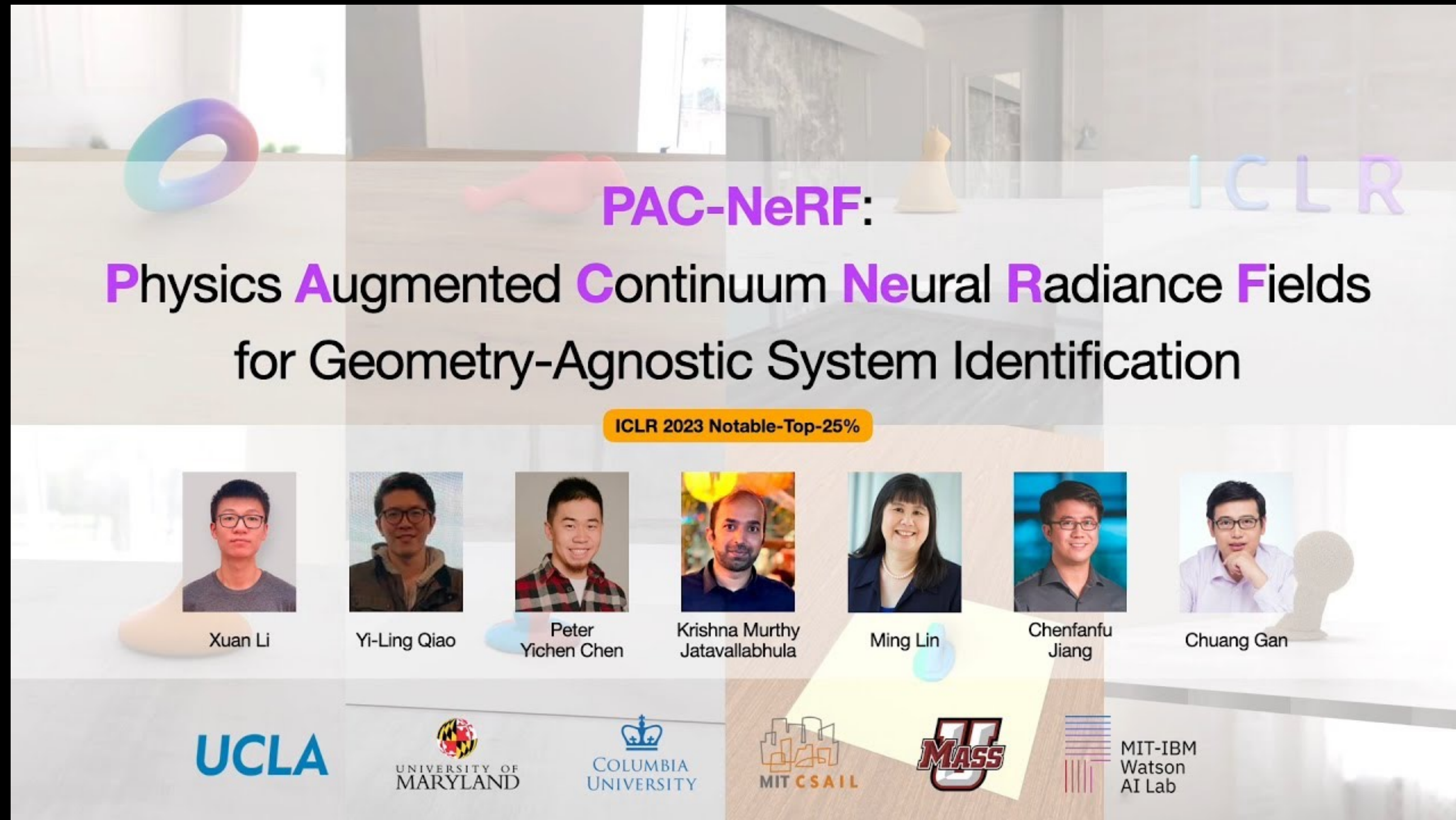


# PAC-NeRF: Model Architecture

|               |                     | Ours                              | Multi-view LSTM                   | D-NeRF<br>+ DiffSim | NeRF<br>+ $\nabla$ Sim | Random           | Oracle ( $\times 10^{-2}$ )       |
|---------------|---------------------|-----------------------------------|-----------------------------------|---------------------|------------------------|------------------|-----------------------------------|
| Newtonian     | $\log_{10}(\mu)$    | <b><math>11.6 \pm 6.60</math></b> | $14.1 \pm 9.26$                   | $> 291.3$           | Not supported          | $64.1 \pm 31.3$  | $19.4 \pm 6.95$                   |
|               | $\log_{10}(\kappa)$ | <b><math>16.7 \pm 5.37</math></b> | $28.2 \pm 19.8$                   | $> 85.2$            |                        | $72.8 \pm 42.5$  | $258.9 \pm 25.2$                  |
|               | $v$                 | <b><math>0.86 \pm 1.45</math></b> | $23.1 \pm 17.8$                   | 25.5                |                        | $55.5 \pm 22.6$  | $2.82 \pm 1.72$                   |
| Non-Newtonian | $\log_{10}(\mu)$    | <b><math>24.1 \pm 21.9</math></b> | $39.4 \pm 26.9$                   | $> 276.7$           | Not supported          | $34.2 \pm 19.9$  | <b><math>24.0 \pm 17.1</math></b> |
|               | $\log_{10}(\kappa)$ | $44.0 \pm 26.3$                   | <b><math>25.1 \pm 22.6</math></b> | $> 262.5$           |                        | $67.6 \pm 49.4$  | $82.90 \pm 21.0$                  |
|               | $\log_{10}(\tau_Y)$ | <b><math>5.09 \pm 7.41</math></b> | $7.19 \pm 7.88$                   | $> 359.6$           |                        | $30.0 \pm 15.0$  | $9.75 \pm 11.3$                   |
|               | $\log_{10}(\eta)$   | <b><math>28.7 \pm 23.3</math></b> | $39.9 \pm 21.1$                   | $> 86.1$            |                        | $64.3 \pm 43.7$  | $90.6 \pm 37.2$                   |
|               | $v$                 | <b><math>0.29 \pm 0.13</math></b> | $24.4 \pm 14.2$                   | 25.9                |                        | $52.2 \pm 21.6$  | $2.60 \pm 0.39$                   |
| Elasticity    | $\log_{10}(E)$      | <b><math>3.02 \pm 3.72</math></b> | $17.7 \pm 9.25$                   | $> 437.5$           | $151.6 \pm 42.6$       | $96.2 \pm 46.9$  | $4.39 \pm 4.54$                   |
|               | $\nu$               | <b><math>4.35 \pm 5.08</math></b> | $81.8 \pm 58.4$                   | 7.67                | $16.4 \pm 6.58$        | $10.9 \pm 6.37$  | $3.65 \pm 2.88$                   |
|               | $v$                 | <b><math>0.50 \pm 0.23</math></b> | $6.10 \pm 3.27$                   | 30.7                | $182.4 \pm 74.1$       | $43.6 \pm 25.4$  | $2.69 \pm 0.97$                   |
| Plasticine    | $\log_{10}(E)$      | $83.8 \pm 68.4$                   | $41.1 \pm 31.4$                   | <b>23.2</b>         | Not supported          | $42.9 \pm 38.0$  | $56.2 \pm 30.7$                   |
|               | $\log_{10}(\tau_Y)$ | <b><math>11.2 \pm 14.5</math></b> | $28.6 \pm 17.3$                   | $> 268.9$           |                        | $82.7 \pm 43.1$  | $8.13 \pm 3.61$                   |
|               | $\nu$               | $18.9 \pm 15.7$                   | <b><math>5.40 \pm 3.49</math></b> | 10.36               |                        | $10.6 \pm 6.15$  | <b><math>4.44 \pm 3.61</math></b> |
|               | $v$                 | <b><math>0.56 \pm 0.17</math></b> | $22.0 \pm 14.8$                   | $> 384.6$           |                        | $47.7 \pm 17.53$ | $4.06 \pm 1.87$                   |
| Sand          | $\theta_{fric}$     | <b><math>4.89 \pm 1.10</math></b> | $20.1 \pm 2.06$                   | 64.5                | Not supported          | $22.5 \pm 14.7$  | <b><math>0.44 \pm 0.42</math></b> |
|               | $v$                 | <b><math>0.21 \pm 0.08</math></b> | $14.6 \pm 6.25$                   | 40.6                |                        | $38.7 \pm 21.1$  | $1.17 \pm 0.42$                   |

# PAC-NeRF: Demo



**PAC-NeRF:**  
Physics Augmented Continuum Neural Radiance Fields  
for Geometry-Agnostic System Identification

ICLR 2023 Notable-Top-25%

Xuan Li   Yi-Ling Qiao   Peter Yichen Chen   Krishna Murthy Jatavallabhula   Ming Lin   Chenfanfu Jiang   Chuang Gan

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